MINISTRY OF EDUCATION AND VOCATIONAL TRAINING
ILBORU SECONDARY SCHOOL
F.IV PRE MOCK 2009

Time :……

SECTION A

Answer all questions

1 (a) Factorize completely

(i) $3x^2 + 7x + 6$

(ii) *evaluate* $(100003)^2 - (99997)^2$

(iii) Express $\frac{2}{\sqrt{7} - \sqrt{3}}$ with a rational denomination

(b) (i) Simplify $\frac{\frac{1}{2.2} + \frac{1}{3.5} + \frac{1}{4.5}}{3\frac{2}{3} \div \frac{0.07}{0.12}}$

(ii) calculate the sum of interior angles of a regular octagon

2. (a) Find the coordinates of the point o n the line presented by 2x-3y+7 = 0 which is equidistant from points p(-4,-80 and Q (7,-1)

(b) Use tables to evaluate $\sqrt{\frac{(.8524)^3 x 24.86}{99.28\% 15.53}}$

3. (a) Solve the equations $2x^3 - 6x + 4x^2y - 2x^2y - 4xy^2 = 0$, and $x - y - 2 = 0$

(b) The area of a regular hexagon inscribed in a circle is $720m^2$, find the radius of the circle

(c) If $f(x) = 4 - 3x - 2x^2$ *find* $f^{-1}(x)$

4. (a) Find the remainder when $f(x) = 5x^3 + 7x^2 - 3x + 2$ is divided by (x – 3)

(b) Kuchi received 4000Tsh. From his father, he spent the money as follows 40% to buy soda, 20% to buy exercise books and 5% of the left to buy sugar in simplified from find the fraction of money left.

(c) Three forces F1=3i + 4j, F2 =5i -2j and F3 = 4i + 3j measured in Newton act at point P (0,0).

(i) Determine the magnitude and direction of their resultant

(ii) Find the magnitude and direction of the opposite of the resultant force.

5. (a) A geometric progression has first term = 2 and second term = 4, find the product of the 1st 100 terms

(b) Find domain and range if the relation

$R = \{(x, y) : y > x^2 + 1 \text{ and } y < 5\}$

6. (a) Use figure below to calculate (i) MN (ii) MY

(b) the angel of elevation of the top of a house 24m high is observed from the top and from the bottom of a vertical ladder, and found to be 45^0 and 60^0 respectively find the height of the ladder.

7. A boat sails 3.202×10^6 dm from O to P on a bearing of 32^0 s and then to Q on a bearing W 50^0S if OQ = 4×10^6dm find (a) the distance from P to Q

(b) the bearing of O from Q

8. (a) If a = 3i+4j, B= 5i – 3j and c = 4i+3j

Find (i) V if V= a+b+c (ii) magnitude of V (iii) unit vector in the direction of V

(b) If $n(A \cap B') = 8\ n(B \cap A') = 5\ and\ n(A \cup B) = 20$

(i) Display the information on a Venn diagram

(ii) Give the values of n (A) and n(B).

9. (a) An area of 24.7cm^2 was plotted on a map of scale 1.50000. what is the area in km^2 on the earths surface.

(b)

10. (a) A vag contains 3 black mables and 2 white mables, a mable is taken from the box and then replaced. A second mable is chosen what is the probability that

(i) They are both black

(ii) One is black and one is white

(iii) The probabilities if the two mables are chosen without replacement

(b) Given that $A = \begin{pmatrix} 4 & -3 \\ 1 & -2 \end{pmatrix}\ B = \begin{pmatrix} 3 & 2 \\ -4 & 0 \end{pmatrix} and\ C = \begin{pmatrix} -1 & 3 \\ 5 & -2 \end{pmatrix}$ find |3A -2B + C|

SECTION B.

11. the following are the scores of 100 students in maths test

Length	40 - 42	43 - 45	46 - 48	49 - 51	52 - 54	55 - 57	58 - 60
freq.	4	9	13	20	34	18	2

(i) Calculate the mean of the distribution by using assumed mean.

(ii) Draw on ogive and use it to estimate the median

(iii)Draw histogram and use it to estimate the mode

12. (a) A slip sails due North from latitude 20^0S for a distance of 1440km. find latitude of the pint it reaches

(b) An aeroplane takes off from B(5^0S, 33^0E) to C(5^0S,39^0E) at a speed of 332km /L. if it leaves B at 3.00p,m at what time will it arrive at C airport? (Circumference of Earth =4 x 10^4 km).

(c) A linear transformation T maps of paint (-1,0) onto (0,1) and the pint (0,-1) onto (-1,0) find (i) the matrix T of transformation

(ii) The image of the pint (2,-1) under T

13. ABCD is a right triangular pyraund with base ABC which is an equilateral triangle given that AB = BC= CA= 8m and DA = DB= DC=10m calculate

(i) The height of the pyraund

(ii) The angle BD makes with the plane ABC

(iii) The angle between the planes DAB and ABC

14. (a) A man requires 15 ments of chemical A and B respectively for his garden . A liquid product contains 5 and 2 units of A and B respectively per jar. A dry product contains 1 and 4 units of A and B respectively per cartin. If a liquid product costs Tsh. 3000/= per jar and the dry product costs sh. 2000/= per carton how many of each should a person purchsase to minimize the cost and meet the requirement

(b) Use matrix increase method to calculate for X and y give that

$$\begin{cases} 3x + y = 17 \\ 27 - 3x - 6y = 0 \end{cases}$$

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